Time Series Imputation — Evaluation Metrics Used

Metrics per Algorithm

54 algorithms across 5 categories. Metrics and experimental setups verified against the primary papers. Metrics are separated by the role they play: training objective, imputation quality, or downstream task.

Algorithm	Metrics Used	Intuition	Setup
Foundation Models			
GPT4TS *multi-task*	**Training:** not stated anywhere in the paper **Imputation:** MSE, MAE **Forecasting:** MSE, MAE (long); sMAPE, MASE, OWA (short); MAPE, ND (zero-shot) **Classification:** Accuracy **Anomaly det.:** F1; P/R/F1 in appendix	A frozen GPT-2 backbone reused for time series, with only the input embedding and task heads fine-tuned. The paper never states what loss any task is trained with, so the reported MSE cannot be checked against an objective.	6 datasets: ETTh1, ETTh2, ETTm1, ETTm2, Electricity, Weather. Point-level random masking at 12.5%, 25%, 37.5%, 50%. States it adheres to the experimental settings of TimesNet.
MOMENT *multi-task*	**Training:** MSE on masked patches **Imputation:** MSE, MAE **Forecasting:** MSE, MAE (long); sMAPE (short) **Classification:** Accuracy **Anomaly det.:** Adjusted best F1, VUS-ROC	Foundation model pretrained by masked patch reconstruction. The only paper in the corpus that criticises another paper's metric choice: it rejects vanilla F1 for anomaly detection as ignoring the sequential nature of time series, then reports MSE and MAE for imputation without comment.	6 datasets: the four ETT sets, Electricity and Weather. Contiguous patch masking (length-8 patches) at 12.5%, 25%, 37.5%, 50%. Anomaly detection on UCR Anomaly archive series; the paper says 44 in the text and 248 in the table.
NuwaTS	**Training:** MSE + InfoNCE contrastive **Imputation:** MSE, MAE (body); RMSE, MAE (one appendix table) **Forecasting:** MSE, MAE	One-for-all pretrained imputer applied across domains without retraining. The appendix metric switch is inherited: that one experiment follows the SAITS evaluation pipeline.	10 datasets (ETT family, ECL, Weather, PEMS). Nine missing rates from 0.1 to 0.9, results averaged across rates. Appendix Table 10 uses Beijing Multi-Site Air-Quality with 10% held out.
Timer	**Training:** MSE, token-wise autoregressive **Imputation:** MSE, reported as win counts over 44 scenarios **Forecasting:** MSE (headline); MAE only in supplementary tables **Anomaly det.:** count of detected anomalies per confidence quantile	Generative pretrained transformer. Imputation results are aggregated as win percentages, so the metric is never reported on its own scale in the main text.	Segment-level masking, deliberately harder than the TimesNet point-level protocol. 44 imputation scenarios across datasets and mask ratios, at data scarcities of 5%, 20% and 100%.
UniTS *multi-task*	**Training:** MSE (masked reconstruction); cross-entropy (classification) **Imputation:** MSE, MAE **Forecasting:** MSE, MAE **Classification:** Accuracy **Anomaly det.:** Precision, Recall, F1	Unified multi-task model with shared weights across tasks. Despite covering the same four tasks as MOMENT, it uses no VUS-ROC and gives no justification for any metric.	4 datasets in the single-task setting: ETTm1, ETTh1, ECL, Weather; the few-shot experiments use six. Point-level random masking at 12.5%, 25%, 37.5%, 50%; few-shot experiments use block-wise masking at 25% and 50%. Baselines: 16 imputation methods, including TimesNet, ETSformer, LightTS, DLinear, FEDformer, Stationary, Autoformer, Pyraformer, Informer, LogTrans, Reformer, LSTM, TCN and LSSL.
Deep Learning			
MissNet	**Training:** joint EM with Viterbi approximation; no point loss **Imputation:** RMSE	State-space model with regime switching and a graphical-lasso sparse network over series. RMSE is cited to DCMF, its closest baseline.	12 datasets: synthetic PatternA and PatternB, MotionCapture (9 motion types), Motes. Missing blocks of 0–5% of series length placed at random until the target rate is reached; Motes also has 9.6% natural missingness including a blackout. Rates 10%–80%. Baselines: Linear, Quadratic, SoftImpute, CDRec, TRMF, DynaMMo, DCMF, BRITS, SAITS, TIDER.
MPIN	**Training:** MSE, two-layer weighted reconstruction **Model selection:** MAE on a held-out validation split **Imputation:** MAE, MRE	Message propagation over a dynamic graph of sensor tuples. One of the few papers that separates the criterion used to pick the model state from the criterion used to report results, though both are MAE.	3 datasets: ICU, Airquality, Wi-Fi. Point-level random removal. Rates 10/30/50% (ICU, Wi-Fi) and 40/60/80% (Airquality). Baselines: MEAN, KNN, MICE, MF, BRITS, SAITS, FP. Metrics justified by citation to BRITS, GRIN and SAITS.
BitGraph	**Training:** masked MAE over forecast targets **Imputation:** none **Forecasting:** MAE, RMSE, MAPE	Forecasting model that absorbs missing inputs. Imputation appears only as a baseline preprocessing step, compared by its downstream forecasting error.	5 datasets: Metr-LA, Electricity, PEMS, ETTh1, BeijingAir. Point-level random dropping at 10/20/40/60/80%; appendix adds block-missing masks generated following GRIN. Baselines: BRITS, SPIN, GRIN, GCN-M, CRU, Transformer, STWA, FEDformer, AGCRN, MTGNN.
BayOTIDE	**Training:** online Bayesian inference **Imputation (point):** RMSE, MAE **Imputation (prob.):** CRPS, NLLK	Bayesian online imputation with functional decomposition into trend and seasonal processes. Reports both point and probabilistic metrics so it can be compared against deterministic baselines.	3 datasets: Traffic-GuangZhou, Solar-Power, Uber-Move. Observed ratios 50% and 70%. 50 posterior samples used for CRPS and NLLK; NLLK results in the appendix.

Algorithm	Metrics Used	Intuition	Setup
TimesNet *multi-task*	**Training:** task-specific (MSE for imputation) **Imputation:** MSE, MAE **Forecasting:** MSE, MAE (long); sMAPE, MASE, OWA (short) **Classification:** Accuracy **Anomaly det.:** Precision, Recall, F1	Reshapes 1D series into 2D by period to apply convolutional blocks. Chooses three scale-free metrics for short-horizon forecasting, following N-BEATS and the M4 competition, and two pointwise metrics for imputation.	6 datasets: ETTm1, ETTm2, ETTh1, ETTh2, Electricity, Weather. Point-level random masking of length-96 series at 12.5%, 25%, 37.5%, 50%. Baselines: ETSformer, LightTS, DLinear, FEDformer, Stationary, Autoformer, Pyraformer, Informer, LogTrans, Reformer.
SAITS	**Training:** MAE (masked imputation + observed reconstruction) **Imputation:** MAE, RMSE, MRE; MSE for the NRTSI comparison **Classification:** ROC-AUC, PR-AUC, F1	Diagonally-masked self-attention with a joint-optimisation training scheme. Adds MSE for one comparison purely because the baseline reported it, the clearest single instance of metric inheritance in the corpus.	PhysioNet-2012, Air-Quality, Electricity and ETT. Classification metrics justified by class imbalance. The MSE tables compare against NRTSI on two further datasets, Air and Gas, which the paper distinguishes from Air-Quality.
PriSTI	**Training:** squared error on predicted noise **Imputation:** MAE, MSE, CRPS **Forecasting (downstream):** MAE, RMSE	Conditional diffusion with a spatiotemporal prior. RMSE appears only for the downstream forecasting experiment.	3 datasets: AQI-36, METR-LA, PEMS-BAY. Two patterns on traffic: block missing (5% random plus 1–4 h sensor failures at 0.15% probability) and point missing (25% random); AQI-36 simulates the real missing distribution. Resulting rates 9.2%–31.1%; sparsity study 10%–90% on METR-LA. Baselines: 15 methods, including MEAN, KNN, Lin-ITP, MICE, VAR, Kalman, TRMF, BRITS, GRIN and CSDI.
GRIN	**Training:** MAE, element-wise, over all stages **Imputation:** MAE, MSE, MRE	Bidirectional graph recurrent network propagating across a sensor network as well as through time. Metrics are named but never defined; MRE is attributed to BRITS. RMSE never appears.	AQI, AQI-36, PEMS-BAY, METR-LA, CER-E. Block missing (5% per sensor per step plus failures at 0.15% probability, lasting 1–4 h or 2 h–2 days) and point missing (25%). Injected rates 8.4%–25%. Baselines: MEAN, KNN, MICE, MF, VAR, rGAIN, BRITS, MPGRU.
CSDI	**Training:** squared error on predicted noise **Imputation:** MAE (deterministic), CRPS (probabilistic) **Interpolation:** CRPS **Forecasting:** CRPS-sum	Conditional score-based diffusion. CRPS is justified as frequently used for probabilistic forecasting and as measuring the compatibility of an estimated distribution with an observation.	PhysioNet-2012 (about 80% natural missingness) and Beijing air quality, 36 stations (about 13%). Ground truth drawn from observed values at 10/50/90%; the paper notes the air-quality missing pattern is not random. 100 posterior samples per missing value. Baselines: Multitask GP, GP-VAE, V-RIN, BRITS, GLIMA, RDIS.
DeepMVI	**Training:** likelihood maximisation, no named loss **Imputation:** MAE	Transformer over a multidimensional array with a kernel-regression fallback. Uses MAE as its sole metric and gives no reason for it; RMSE is mentioned once in the paper, in passing.	10 structurally heterogeneous datasets. Running time reported separately as efficiency; memory is not measured.
HKMF-T	**Training:** L_2 estimation error + temporal smoothness + tag similarity + regulariser, by SGD **Imputation:** average RMSE, average DTW distance	Hankel matrix factorisation with side-information tags, designed for blackouts. The only paper in the corpus that reports a temporal metric its objective does not contain, though it justifies DTW only as a frequently used elastic distance measure.	3 datasets: Bike Sharing, Motor Vehicle Crash, Electric Power Consumption. Blackouts evaluated by leave-one-segment-out cross validation, sliding to full coverage; blackout length 1–20 steps. Baselines: DynaMMo, linear interpolation, MA Tag, Tag Mean.
SSGAN	**Training:** MAE reconstruction + adversarial + cross-entropy classifier **Imputation:** RMSE (body), MAE (appendix) **Downstream:** AUC, RMSE	Semi-supervised GAN using label information to guide imputation. MAE is relegated to the appendix because it shows similar trends to RMSE and space is limited; the paper gives RMSE no methodological preference.	PhysioNet, KDD and Activity. Post-imputation classification scored by AUC, regression by RMSE.
GP-VAE	**Training:** ELBO on observed features **Imputation:** NLL, MSE **Downstream:** AUROC	Gaussian-process prior in the latent space of a VAE. The only paper in the corpus that validates its proxy metric, reporting that downstream classification correlates with the ground-truth likelihood where ground truth exists.	Healing MNIST and SPRITES for reconstruction; PhysioNet for the downstream task, chosen because roughly 80% of features are missing and direct evaluation is impossible. Missingness-mechanism study covers Random, Spatial, Temporal+, Temporal− and MNAR. Logistic regression used downstream, for interpretability.
NAOMI	**Training:** MSE (deterministic); adversarial (stochastic) **Imputation:** L2 loss **Basketball:** five domain-specific trajectory measures	Non-autoregressive multiresolution imputation. Argues explicitly that an L2 loss is not a good indicator of realistic trajectories in a stochastic environment, and replaces it with domain measures. The only paper in the corpus that questions whether a metric fits what is being reconstructed.	3 datasets: PEMS-SF traffic, simulated billiards, basketball player tracking. A random number of steps is masked, then the specific steps are sampled at random. Rates given as counts: 122–140 of 144 (traffic), 180–195 of 200 (billiards), 40–49 of 50 (basketball). Baselines: Linear, KNN, GRUI, MaskGAN, SingleRes.
M-RNN	**Training:** MSE on observed values **Imputation:** RMSE **Downstream:** AUROC	Multi-directional recurrent network interpolating within streams and imputing across them. RMSE is justified only as “common”; all measurements are scaled to $[0, 1]$, so the reported values are not in clinical units.	5 medical datasets, including MIMIC-III and the UNOS heart and lung transplantation data, with 5-fold cross-validation. The downstream task differs per dataset: 24-hour mortality on MIMIC-III, one-year mortality on UNOS.
BRITS	**Training:** MAE reconstruction + MAE consistency + downstream output loss **Imputation:** MAE, MRE **Downstream:** AUC (health-care), Accuracy (activity)	Bidirectional recurrent imputation treating missing values as variables of the graph. MRE is computed as a ratio of totals, although the name says mean. The downstream metric is chosen by class balance, stated explicitly.	Health-care, activity and air-quality datasets. The first two are evaluated on normalised data because the numerical scales differ; the air-quality experiment is not normalised.

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GAIN	**Training:** adversarial + reconstruction **Imputation:** RMSE **Downstream:** AUROC	Generative adversarial imputation with a hint mechanism. Tabular in origin.	5 UCI datasets: Breast, Spam, Letter, Credit, News. MCAR point-level removal, 20% by default, varied from 10% to 90% on Credit. Baselines: MICE, MissForest, matrix completion, auto-encoder, EM.
WGAN (GRUI-GAN)	**Training:** WGAN losses + masked squared reconstruction **Imputation:** MSE **Downstream:** AUC, MSE	GRU-based generator with a time-decay mechanism, imputing by optimising the input noise vector. States plainly that direct evaluation is impossible without a complete dataset, which is why the downstream task is used.	PhysioNet (no ground truth, downstream only) and KDD (direct MSE possible). AUC chosen for the imbalanced mortality labels.
E^{2}GAN	**Training:** squared error + adversarial **Imputation:** MSE **Downstream:** AUC, MSE	End-to-end successor to GRUI-GAN using an autoencoder generator. Gives both the no-ground-truth reason and the class-imbalance reason in a single passage.	PhysioNet (13.85% positive labels) and KDD. Runtime reported separately.
SSSD	**Training:** MSE on predicted noise **Imputation:** MAE, RMSE, MSE, MRE **Forecasting:** MAE, RMSE (PTB-XL); MSE (Solar); MAE, MSE (ETTm1)	Diffusion with a structured state-space backbone. Reports different metric subsets per dataset, stating that it uses the commonly used metrics for each, and gives per-metric error-sensitivity reasoning in its appendix.	PTB-XL, as ECG (MAE, RMSE); MuJoCo (MSE); Electricity (MAE, RMSE, MRE); PEMS-BAY and METR-LA (MAE, MSE, MRE); ETTm1 and Solar for forecasting.
MTSIT	**Training:** MSE on stochastically masked positions **Imputation:** MAE	Transformer autoencoder trained by geometric block masking. Trains on a squared error and reports an absolute one.	PhysioNet-2012 (about 80% missing) and Beijing air quality, 36 stations (about 13%). Training masks segments of geometric-distributed length; test ground truth masked at 10/50/90% of observed values. Baselines: V-RIN, BRITS, RDIS, CSDI, unconditional CSDI, GLIMA, SAITS.
TIDER	**Training:** masked reconstruction + smoothness + AR + regularisation **Imputation:** RMSE, MAE, MAPE	Matrix factorisation into disentangled trend, seasonality and local bias components.	3 main datasets: Guangzhou traffic, Solar-Energy, Westminster Uber Movement; appendix adds Household Power and Beijing Climate. Entries removed at random, no block pattern stated. Rates 20% and 40%. Baselines: SimpleMean, KNN, MICE, MF, MF+L2, SoftImpute, TRMF, BRITS, SAITS, GAIN, NAOMI, SingleRes, CSDI.
PoGeVon	**Training:** ELBO with MAE reconstruction + Frobenius link term + KL **Imputation:** MAE, MSE, MRE **Link prediction:** Frobenius norm	Position-aware graph enhanced VAE for networked series, imputing node features and predicting links jointly.	5 datasets: COVID-19, AQ36, PeMS-BA, PeMS-LA, PeMS-SD. Random node-feature masking plus edge masking when an end node is missing; AQ36 follows the simulated-real-distribution protocol. Feature rates 25% (13.24% on AQ36); edge rates 28%–44%; sensitivity sweep 15%–45%. Baselines: Mean, MF, MICE, BRITS, rGAIN, GRIN, NET3.
CSBI	**Training:** Schrödinger bridge, forward-backward SDE iterations **Imputation:** RMSE, MAE, CRPS **Forecasting:** CRPS	Probabilistic imputation via a provably convergent Schrödinger bridge. CRPS is described as measuring the quality of the imputed distribution.	Synthetic data, PM2.5 air quality (13% natural missingness) and PhysioNet-2012 (80%). Synthetic targets are 20 consecutive time points per feature; PhysioNet targets randomly selected from observed values. Artificial target rates 10% and 50%. Baselines: V-RIN, Multitask GP, GP-VAE, CSDI.
TimeCIB	**Training:** ELBO + KL + temporal contrastive term **Imputation:** NLL, MSE **Classification:** AUROC **Forecasting:** ForecastMSE	Conditional information bottleneck applied to imputation. Restricts NLL to VAE-based baselines and marks the restriction in its tables, giving three specific reasons why other methods are excluded.	5 datasets: HealingMNIST, RotatedMNIST, Beijing Air Quality, US Local, PhysioNet-2012. HealingMNIST uses MNAR pixel missingness; RotatedMNIST removes all features at one time step. Rates 60% artificial, PhysioNet 80% natural plus 2% artificial; robustness sweep 10%–90% across Random, Spatial, Temporal+ and Temporal−. Baselines: GP-VAE, HI-VAE, BRITS, SAITS, mTANs, CSDI.
FGTI	**Training:** MSE on predicted noise **Imputation:** RMSE, MAE; CRPS for the probabilistic baselines (appendix) **Downstream:** RMSE (air quality), AUC (mortality)	Frequency-guided diffusion separating high-frequency from dominant-frequency components, the latter selected by largest amplitude. Justifies RMSE and MAE only as following previous studies, but gives CRPS a purpose: measuring the gap between the learned and the true distribution.	3 datasets: KDD Beijing air quality (4.46% natural), Guangzhou traffic (1.29%), PhysioNet (79.71%). MCAR removal at 10/20/30/40%, with MAR and MNAR also varied; ablation covers block, mix and random masks. Baselines: 15 methods, including Mean, BTMF, TIDER, BRITS, TST, SAITS, TimesNet, GRIN, CSDI, SSSD and PriSTI.
CATSI	**Training:** sum of three mean squared deviations on observed entries **Imputation:** nRMSD	Context-aware recurrent imputation for clinical analytes, combining a patient-level context vector with cross-feature imputation. Normalises by the per-patient per-analyte range, and calls the result a deviation rather than an error.	DACMI challenge data, 13 clinical analytes. Individual and consecutive missingness evaluated separately. No justification given for the normalisation.
CSAI	**Training:** MAE reconstruction + MAE consistency (+ BCE end-to-end) **Imputation:** MAE **Classification:** AUC / ROC-AUC	Knowledge-enhanced conditional imputation for electronic health records, with a non-uniform masking strategy. No justification given for either metric.	eICU, MIMIC-59 and PhysioNet at 5%, 10% and 20% masking. Compute cost reported: best epoch, parameters, peak GPU memory, training and inference time.
SST-GCN	**Training:** KL divergence between estimated and true speed distributions **Imputation:** KLD, JSD, Wasserstein distance (reported as EMD)	Imputes a distribution over traffic speed rather than a point estimate. The only paper in the corpus reporting three distributional divergences, and it says why: standard regression metrics are not applicable to distributions. Normalises the divergences because KLD and EMD are unbounded.	2 datasets: Highway Tollgates Network and City Road Network (subgraphs CR173, CR1026). A random subset of *nodes* has its speed distribution removed, at 50/60/70/80% of nodes; appendix adds spatially localised missingness by BFS clustering. Baselines: GMAN, TCN, ST-MetaNet, GWaveNet, MTGNN, GCWC, VAR, MICE.

Algorithm	Metrics Used	Intuition	Setup
FIM-ℓ	**Training:** supervised pretraining on synthetic ODE solutions; likelihood plus a one-step reconstruction term **Imputation:** MAE (main); RMSE, MRE, R^2 (appendix)	Zero-shot imputation by a foundation inference model, applied without fine-tuning on the target domain. R^2 is reported for baseline compatibility.	Metrics computed only at the missing points, following the methodology of its baselines. ODEBench used for the R^2 comparison, where no targets are masked.
ReCTSi	**Training:** masked MAE over missing regions **Imputation:** MAE, MSE, MRE	Resource-efficient imputation with decoupled pattern learning. One of two papers in the corpus that give a separate rationale for each metric they report, the other being LSCD.	5 datasets: PeMS-BA, PeMS-LA, PeMS-SD, AQ36, COVID-19, with dataset missing rates of 25% (13% for AQ36). The masking protocol is not stated; the paper says its configuration follows PoGeVon. Efficiency reported alongside accuracy: FLOPs, parameters, latency, peak memory. Baselines: MEAN, MF, MICE, rGAIN, PoGeVon, BRITS, TimesNet, GRIN, SAITS, NET3.
SSD-TS	**Training:** noise-prediction norm (DDPM) **Imputation (det.):** MAE, MSE, RMSE, MRE **Imputation (prob.):** CRPS **Forecasting:** MAE, MSE	State-space diffusion. Explicitly splits evaluation by whether a model can quantify uncertainty, adding CRPS only for probabilistic models.	MuJoCo (MSE), PhysioNet and AQI (RMSE), PTB-XL and Solar in the appendix, ETTm1 for forecasting. CRPS reported only for probabilistic baselines.
LSCD	**Training:** noise-prediction MSE + Lomb-Scargle spectral consistency **Imputation:** MAE, RMSE, spectral MAE	Diffusion conditioned on the Lomb-Scargle periodogram. The only paper in the corpus reporting a spectral metric, and the only one to argue for combining time-domain and frequency-domain measures to assess both value accuracy and spectral preservation.	Spectral MAE computed between normalised power spectral densities, with missing values masked. The introduction claims a CRPS improvement; the word appears once in the paper and no CRPS value is reported anywhere.
DMM	**Training:** ELBO with reconstruction + KL over latent states and missing-cause variables **Imputation:** MAE, MSE **Simulation:** MCC	Causal treatment of the missingness mechanism, with identification results for when the mechanism can be recovered.	Simulation plus 6 real datasets: ETTh1, ETTh2, ETTm1, ETTm2, Exchange, Weather; appendix adds MIMIC-III. Masks generated per mechanism, MAR and MNAR in the main experiments, mixed settings in the appendix. Mask ratios 0.2, 0.4, 0.6. Baselines: SAITS, ImputeFormer, CSDI, TimesNet, BRITS, SSGAN, TimeCIB, GP-VAE.
Matrix Completion			
CDRec	**Objective:** centroid decomposition of the interpolation-filled matrix, written back only at the missing positions **Imputation:** RMSE	Memory-efficient recovery of long missing blocks in correlated series via centroid decomposition, iterating until the normalised Frobenius change falls below a threshold.	5 datasets: BAFU, MeteoSwiss, Baseball, Temperature, Gas. Contiguous blocks: one block in the middle of one series, or blocks placed at random across several series. Block sizes 20/40/60/80% of the chosen series, 10% for the multi-series experiments; 2–4 incomplete series. Baselines: SSA, GROUSE, TeNMF, TKCM, ST-MVL, M-RNN, TRMF.
TRMF	**Objective:** regularised matrix factorisation with a temporal autoregressive term **Imputation:** ND, NRMSE **Forecasting:** ND, NRMSE	Temporal regularised matrix factorisation, primarily a forecasting method with imputation as a secondary task. Both metric definitions are deferred to an appendix that is not part of the published PDF, so its denominators cannot be checked.	5 datasets: synthetic, electricity, traffic, walmart-1, walmart-2; imputation experiments use synthetic, electricity and traffic. Contiguous occluded blocks of length 2 (synthetic) or 5 (real), placed uniformly at random. Observed ratios 20/30/40/50%. Baselines: TCF, MF, DLM, Mean.
GROUSE	**Objective:** gradient descent on the Grassmannian **Reported:** determinant similarity ζ, Frobenius discrepancy ε **Imputation:** none	Convergence analysis of a subspace-tracking algorithm. Analyses the fully observed case and defers the undersampled case to future work, so it contains no missing-data model and reports no imputation accuracy.	Not an imputation experiment. Both quantities measure how close the estimated subspace is to the true one.
ROSL	**Objective:** row-1 norm of the coefficients plus ℓ_1 norm of the error, subject to an orthonormal dictionary **Imputation:** none (MAE reported for low-rank recovery)	Robust orthonormal subspace learning, recovering a low-rank matrix from sparse corruption. The word "missing" appears once in the paper, inside a reference title.	Not an imputation experiment.
SoftImpute	**Objective:** Frobenius fit + nuclear norm regularisation **Reported:** standardised training and test error; RMSE on Netflix	Iterative soft-thresholded SVD for large incomplete matrices. Separates fitting error on observed entries from prediction error on held-out entries, which probes the regularisation trade-off directly.	Simulated matrices for the train/test error curves; the Netflix Prize data for the RMSE comparison.
SVT	**Objective:** nuclear norm minimisation subject to observed entries **Reported:** relative Frobenius error	Singular value thresholding, formulated as a mathematical recovery algorithm. The stopping criterion is computed on the sampled entries and the reported error on the full matrix, and the paper argues heuristically that the two are of the same order.	Synthetic low-rank matrices of varying size and rank.
SPIRIT	**Objective:** minimise cumulative squared reconstruction error (implicit, incremental) **Reported:** normalised MSE rate; energy ratio as a control parameter **Imputation:** none computed	Streaming PCA over co-evolving series. Justifies its normalised error rate on the grounds that it wants a measure of overall quality weighting every observation equally.	Chlorine, Critter and Motes. The missing-value experiment blanks 1.5% of values in five consecutive blocks and is assessed by eye against a plot; no number is computed on the blanked values.

Algorithm	Metrics Used	Intuition	Setup
IterativeSVD	**Training:** iterative convergence, no explicit loss **Imputation:** nRMSE	Iterative SVD-based estimation of missing values, developed for DNA microarray data. Normalises the RMS error by the average data value in the complete dataset, and states the reason: to allow comparison of estimation accuracy between different datasets.	Gene expression matrices with artificially removed entries.
Pattern Search			
TKCM	**Objective:** L_2 pattern dissimilarity across k anchor points **Imputation:** RMSE	Continuous imputation in streams. The pattern is built from a separate reference series, and the value taken is the incomplete series at the matched anchor points, which are time-aligned with the gap.	4 datasets: SBR (South Tyrol meteorological), SBR-1d, Flights, Chlorine. Contiguous blocks of consecutively missing values, from one week up to six weeks, or 10%–80% of dataset size on Chlorine. Final comparison uses one week (SBR, SBR-1d) and 20% of dataset size (Flights, Chlorine). Baselines: CD, MUSCLES, SPIRIT.
ST-MVL	**Objective:** linear least squares over four views **Imputation:** MAE, MRE	Combines spatial and temporal views for geo-sensory data. MRE is computed as a ratio of totals, the same formulation BRITS uses.	Beijing air quality (36 stations) and meteorology (16 sensors), one year, 8 759 timestamps, four properties. Uses the real missing pattern, stating that real missingness is not random. Rates 13.3%–30.3% depending on the property. Baselines: ARMA, SARIMA, stKNN, Kriging, AKE, DESM, IDW+SES, CF, NMF, NMF-MVL.
DynaMMo	**Objective:** EM maximising expected log-likelihood **Imputation:** MSE in the text, captioned “Average rmse” in the figure **Compression:** RMSE	Linear dynamical system over co-evolving sequences, also used for compression and summarisation. The definition in the text and the figure label disagree on the same page.	2 datasets: Chlorine (166 nodes, 4 310 ticks) and Motion (58 mocap motions, 93 features). Contiguous occlusions of Poisson-distributed duration, repeated until the target fraction is reached. About 10% occluded, average occlusion length varied from 10 to 100. Baselines: linear interpolation, spline interpolation, MSVD.
Machine Learning			
IIM	**Objective:** ridge regression, one model per tuple with adaptive nearest neighbours **Imputation:** RMSE	Learns an individual regression model for each incomplete tuple from its learning neighbours.	9 datasets: ASF, CCS, CCPP, SN, PHASE, CA, DA, MAM, HEP. Point-level random removal of one attribute from 5% of tuples, plus a contiguous variant with clustered incomplete tuples. Baselines: kNN, kNNE, IFC, GMM, SVD, ILLS, GLR, LOESS, BLR, ERACER, PMM, XGB.
XGBoost	**Objective:** regularised gradient boosting **Imputation:** none **Downstream:** AUC, NDCG@10	Handles sparsity by routing missing values to a learned default direction at each split. The word “imputation” does not appear in the paper.	4 datasets: Allstate, Higgs Boson, Yahoo LTRC, Criteo. Accuracy reported on Higgs Boson (AUC) and Yahoo LTRC (NDCG@10); Allstate and Criteo are used for the sparsity and scaling experiments. Not an imputation experiment.
MICE	**Training:** none; chained-equation sampling **Imputation:** none **Criterion:** inferential validity under MAR	Multiple imputation by chained equations. Its criterion is less biased parameter estimation with correct standard errors under Rubin’s rules. RMSE, MAE and MSE appear zero times in the paper.	Not a reconstruction experiment. Results reported as regression coefficients with confidence intervals.
MissForest	**Training:** iterative convergence monitored by change in the imputed values **Imputation (continuous):** NRMSE **Imputation (categorical):** PFC	Non-parametric imputation by random forest for mixed-type data. Normalises by the variance of the true values inside the square root, so its NRMSE is an error relative to the data’s own standard deviation. PFC is defined only for categorical variables.	10 datasets spanning continuous (Isoprenoid gene network, Parkinson’s, Musk, Insulin), categorical (SPECT, Promoter, Lymphography) and mixed-type (Gaucher proteomics, GFOP peptide, Children’s Hospital). Values removed completely at random at 10%, 20% and 30%, over 50 simulations per experiment, reduced to 10 on Insulin for compute reasons. Baselines: KNNimpute, a dummy-coded KNNimpute, MissPALasso, MICE.